

Question				Marking details	Marks Available					
					AO1	AO2	AO3	Total	Maths	Prac
7	(a)			[Flemings] LH rule identified (1) or inferred First finger (N→S) field and second finger (L→R or + → -) current (1) Thumb points up so wire moves up (1) so student correct Full marks can only be awarded if the conclusion is present.			3	3		3
	(b)	(i)		40×10^{-3} or 0.04 [N] (1) Any value in the range 1.75 - 1.8 [A] (1)		2		2		2
		(ii)		Force increases as current increases (1) at a constant rate (1) Alternative: Force is [directly] proportional to current or if current doubles then force doubles (2)		2		2	2	2
		(iii)		Gradient = $B l$ (1) Intercept = 0 (1)		2		2	2	2
		(iv)		Gradient e.g. = $\frac{90 [\times 10^{-3}]}{4} = 22.5 [\times 10^{-3}]$ (1) $B = \frac{22.5 \text{ecf}}{5} [\times 10^{-1}]$ (1) $B = 0.45$ [T] (1) N.B. Tolerance on graph readings i.e. force $\pm 1 [\times 10^{-3}]$ N and current ± 0.05 [A] Answer of 4.5×10^n where n is not equal to -1 award 2 marks Alternative: Selection and manipulation of $B = \frac{F}{Il}$ (1) Substitution into rearranged equation: $B = \frac{90 [\times 10^{-3}]}{4 \times 5 [\times 10^{-2}]} [\times 10^{-1}]$ (1) $B = 0.45$ [T] (1)			3	3	2	2
				Question 7 total	0	6	6	12	6	11